

Solution for CS5275 Problem Set 2

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Note: This solution file is not final. It is currently only a first draft. It has my solutions, but they have not been proofread or revised. It also does not have other parts, like common mistakes, etc. The main point is for you to reference during the midterm exam. As usual, I will upload a revised file at the end of the tutorial discussing this problem set.

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1 Problem 1

Consider the following procedure. Starting from an n -vertex graph $G = (V, E)$, iteratively do the following: as long as there exists a component C and a cut $(S, C \setminus S)$ such that $\Phi_C(S) \leq \phi$, erase the edges in the cut, thus splitting C into two parts (each of one or more components). The procedure stops when there is no more S that can be found.

Prove that the total number of edges erased is

$$O(\phi \cdot |E| \cdot \log n).$$

1.1 My solution

In this solution, we will use a slight abuse of notation. If A, B are subsets of the vertex set V , then $\text{vol}_A(B) = \text{vol}_{G[A]}(B)$ is the volume of B in the induced subgraph $G[A]$.

We append the following to the procedure. We are going to put “pebbles” on vertices. A vertex can have any number of pebbles, and at the beginning, there is no pebble anywhere. The “weight” of a pebble on vertex v is $w(v) := \text{vol}_G(\{v\})$.

Whenever we make a cut $(S, C \setminus S)$, put a pebble on every vertex on the smaller side. In other words, suppose $\text{vol}_C(S) \leq \text{vol}_C(C \setminus S)$ – otherwise we take $S \leftarrow C \setminus S$ – then put a pebble on every vertex in S .

We make the following two claims:

Lemma 1.1. *Consider any step of the procedure, suppose it makes the cut $(S, C \setminus S)$, and suppose S is the smaller side. Then*

$$|E(S, C \setminus S)| \leq \phi \cdot \sum_{v \in S} w(v).$$

In other words, the number of edges cut is at most ϕ times the total weight of pebbles put with this cut.

Lemma 1.2. *At the end of the procedure, each vertex has $\leq 2 \log_2 n$ pebbles.*

We first prove these two lemmas, then complete the proof using these lemmas.

Proof of Lemma 1.1. This is straightforward from the definition of conductance Φ . By definition of the procedure, we have

$$\Phi_C(S) \leq \phi \quad \implies \quad \frac{|E(S, C \setminus S)|}{\min\{\text{vol}_C(S), \text{vol}_C(C \setminus S)\}} \leq \phi$$

Since S is the smaller side,

$$|E(S, C \setminus S)| \leq \phi \cdot \text{vol}_C(S).$$

Finally, $\text{vol}_C(S) \leq \text{vol}_G(S)$ because C is a subgraph of G , and $\text{vol}_G(S) = \sum_{v \in S} \text{vol}_G(\{v\})$. But note that $\text{vol}_G(\{v\})$ is the weight of a pebble on v , so this sum is W . This proves the claim. $\square$

Proof of Lemma 1.2. We claim the following. Suppose a vertex v is in a component C . At some step, C is cut so that v now belongs in a smaller component S . Then if v gets a pebble, then

$$\text{vol}_S(S) \leq \frac{\text{vol}_C(C)}{2}.$$

In other words, whenever a vertex gets pebbled, its component has been halved.

But this is straightforward. We have $\text{vol}_S(S) \leq \text{vol}_C(S)$, and if S is pebbled, that means $\text{vol}_C(S) \leq \text{vol}_C(C \setminus S)$. Therefore $\text{vol}_C(S) \leq \text{vol}_C(C)/2$.

Now, $\text{vol}_V(V) \leq n^2$ since G is unweighted. Therefore a vertex v can only be pebbled up to $2 \log_2 n$ times before its component has volume < 1 . When this happens, the component has volume 0, i.e. v is an isolated vertex. Then it will not be involved in any future cut. This proves the claim. $\square$

We now return to the main proof. Suppose the procedure has finished. Let $E^\times$ be the set of removed edges, and define p_v to be the number of pebbles on vertex v . By Lemma 1.1 summed over all cuts,

$$|E^\times| \leq \phi \cdot \sum_v (p_v \cdot w(v)).$$

By Lemma 1.2, $p_v \leq 2 \log n$. Also recall $w(v) = \text{vol}_G(\{v\})$. Therefore

$$\begin{aligned} |E^\times| &\leq \phi \cdot \sum_v (p_v \cdot w(v)) \\ &\leq \phi \cdot 2 \log_2 n \cdot \sum_v w(v) \\ &= \phi \cdot 2 \log_2 n \cdot \text{vol}_G(V) \\ &= \phi \cdot 2 \log_2 n \cdot 2 \cdot |E| \\ &= O(\phi \cdot |E| \cdot \log n) \end{aligned}$$

1.2 Insight and discussion

The core result is Lemma 1.2: a vertex only “contributes” to a cut at most $O(\log n)$ times. The insight is pretty straightforward when spelled out: the main contribution to a cut size comes from the volume of the smaller side, so whenever a vertex contributes to a cut, its side has been halved. The graph itself only has $O(n^2)$ volume, so you can only halve $O(\log n)$ times before the volume becomes < 1 .

The phrasing of the proof by using pebbles is how the solution literally came to my mind. You can write it more “mathematically”, like counting how many times a vertex “contributes” to the upper bound sum, but this is easier to visualize.

2 Problem 2

In this problem, let G be a graph with normalized Laplacian $\mathbf{N}$.

- (a) For every n , construct a connected graph G with n vertices, such that

$$\lambda_2(\mathbf{N}) = O(1/n^3).$$

- (b) Prove that any connected graph G on n vertices satisfies

$$\lambda_2(\mathbf{N}) = \Omega(1/n^3).$$

2.1 My solution

For simplicity, I will use λ_2 to mean $\lambda_2(\mathbf{N})$. Fix a graph $G = (V, E)$, and let $f : V \rightarrow \mathbb{R}$. Write

$$\lambda_2(f) = \frac{\sum_{uv \in E} (f(u) - f(v))^2}{\sum_v \deg(v) f(v)^2}$$

Then $\lambda_2 = \min_f \lambda_2(f)$, among all f satisfying $\sum_v \deg(v) f(v) = 0$ and $f \neq 0$.

Part (a). We will use the “thin bridge” construction (previously used in Problem set 1, Problem 4, as an appendix). For simplicity, let $n = 6k - 1$. The graph G has the following form:

Take $2k$ vertices into a clique, and call it the “positive island” C_+ . Take $2k$ more vertices into another clique, and call it the “negative island” C_- . Then take an arbitrary vertex in C_+ and label it v_{+k} , and take another arbitrary vertex in C_- and label it v_{-k} . Finally, connect v_{+k} and v_{-k} with a path of length $2k$ edges, with vertices $v_{-k}, v_{-k+1}, v_{-k+2}, \dots, v_{+k}$.

We also define the following function f :

$$f(v) = \begin{cases} i & \text{if } v = v_i \text{ is on the thin bridge} \\ k & \text{if } v \text{ is on the positive island } C_+ \\ -k & \text{if } v \text{ is on the negative island } C_- \end{cases}$$

We claim this f satisfies the conditions and so counts for λ_2 , and $\lambda_2(f) = O(k^{-3}) = O(n^{-3})$. Therefore $\lambda_2 \leq \lambda_2(f) = O(n^{-3})$, proving the problem statement.

To see that $\sum_v \deg(v)f(v) = 0$, just note that f is basically anti-symmetrical across v_0 . Also, clearly $f \not\equiv 0$. So this f counts for λ_2 . Now we compute the numerator and the denominator of $\lambda_2(f)$.

Numerator. Most of the edges uv have $f(u) - f(v) = 0$. The only edges that don't are the edges on the thin bridge, which have $(f(u) - f(v))^2 = 1$. There are $2k$ such edges, so the numerator is $2k$.

Denominator. Consider the positive island C_+ by itself. There are $2k$ vertices here. Each vertex here has degree $\Omega(k)$ (specifically, $2k - 1$, except for v_{+k} which has degree $2k$). So the positive island contributes

$$\sum_{v \in C_+} \deg(v)f(v)^2 = 2k \cdot \Omega(k) \cdot k^2 = \Omega(k^4)$$

to the denominator. Therefore the denominator is $\Omega(k^4)$. (Indeed, the negative island also contributes $\Omega(k^4)$; the thin bridge only contributes $\Omega(k^3)$.)

Therefore

$$\lambda_2 \leq \lambda_2(f) = \frac{2k}{\Omega(k^4)} = O(k^{-3}) = O(n^{-3}).$$

Handling other n . The above construction works for $n = 6k - 1$. To make it work for other n , simply attach the additional vertices to the center with $f = 0$. This doesn't change any of the calculations, and the asymptotic is still correct since $k = \Theta(n)$ still.

Part (b). Let f be any function satisfying the conditions of the problem. We will show $\lambda_2(f) = \Omega(n^{-3})$. For this problem, recall that G is connected, and we assume G has ≥ 2 vertices.

First, let $M = \max |f(v)|$ be the maximum absolute value of f . Without loss of generality, suppose M is achieved positively, i.e. there is some $v_{\max}$ such that $f(v_{\max}) = M$, otherwise negate all values of f .

Furthermore, consider the condition $\sum_v \deg(v)f(v) = 0$. Now $\deg(v) > 0$ for all v , so there must be some $v_{\min}$ for which $f(v_{\min}) \leq 0$, otherwise the sum is strictly positive.

Now we look at the numerator and the denominator of $\lambda_2(f)$.

Numerator. There is a path from $v_{\min}$ to $v_{\max}$ because G is connected; label the vertices of this path as $v_0, v_1, v_2, \dots, v_k$ where $v_0 = v_{\min}$ and $v_k = v_{\max}$. Note that $k \leq n$ because there are only n vertices in the graph.

We consider the contribution of this path to the numerator. Using the [QM-AM inequality](#)¹:

$$\frac{\sum_{i=0}^{k-1} (v_i - v_{i+1})^2}{k} \geq \left(\frac{\sum_{i=0}^{k-1} |v_i - v_{i+1}|}{k} \right)^2$$

Then by using the triangle inequality:

$$\sum_{i=0}^{k-1} |v_i - v_{i+1}| \geq |v_k - v_0| = |v_{\max} - v_{\min}| \geq M$$

¹also power mean inequality with exponents 2 and 1, and can be proven by Cauchy-Schwarz

Therefore, the numerator is

$$\geq \sum_{i=0}^{k-1} (v_i - v_{i+1})^2 \geq \frac{1}{k} \cdot \left(\sum_{i=0}^{k-1} |v_i - v_{i+1}| \right)^2 \geq \frac{1}{n} \cdot M^2 = \Omega(M^2 n^{-1})$$

Denominator. There are n vertices, each of which has $\deg(v) = O(n)$. Moreover, by definition of M , we have $f(v)^2 \leq M^2$. Therefore, the denominator is

$$\sum_v \deg(v) f(v)^2 \leq n \cdot O(n) \cdot O(M^2) = O(M^2 n^2)$$

Therefore,

$$\lambda_2(f) = \frac{\Omega(M^2 n^{-1})}{O(M^2 n^2)} = \Omega(n^{-3}).$$

This is true for all f . Therefore

$$\lambda_2 = \min_f \lambda_2(f) = \Omega(n^{-3}).$$

2.2 Insight and discussion

Both parts of the problem stem from the same observation: in order for λ_2 to have a low result, we want $(f(u) - f(v))^2$ to be small all the time, but $f(v)^2$ to be big. The problem is that these two objectives are opposing each other.

The main observation is the QM-AM inequality. You can motivate it from the following. The condition requires there to be some $v_{\min}$ with $f(v_{\min}) \leq 0$. Now you have a $v_{\max}$ with large (positive) $f(v_{\max})$. You have a path from $v_{\min}$ to $v_{\max}$, so intuitively, the best way to keep $\sum (f(u) - f(v))^2$ low is by making gradual incremental steps; each step isn't too large by itself, but the sum over all steps gets you all the way from $f(v_{\min})$ to $f(v_{\max})$. And to formalize this, you can use the QM-AM inequality. There are equivalent formulations, such as using Cauchy-Schwarz. If you start with this observation, you will likely solve part (b) first.

Now you know, for part (a), you need all the inequalities to be tight. For example, the numerator is $\Omega(M^2 n^{-1})$, so you need this to be actually $\Theta(M^2 n^{-1})$. This can happen by having a $\Theta(n)$ -long path that gets you from 0 to M . On the other hand, the denominator is $O(M^2 n^2)$. So you need this to be $\Theta(M^2 n^2)$, by having $\Theta(n)$ vertices at degree $\Theta(n)$ that all have $f = M$. This can happen by having a $\Theta(n)$ -big clique with $f = M$. And piecing this together gives you something like the thin bridge construction.

So, while the construction isn't very natural to come by, writing down all the constraints you have to satisfy can lead you there.

3 Problem 3

Consider the lazy random walk on a graph G and its mixing time $T_{\text{mix}}(G, \varepsilon)$.

Prove that there exists some distribution $\mathbf{x}$ supported on a single vertex (i.e. $\mathbf{x}_v = 1$ for some vertex v , $\mathbf{x}_u = 0$ for all other u), such that it achieves the maximum in the definition of $T_{\text{mix}}(G, \varepsilon)$.

3.1 My solution

We will generalize the random walk slightly. Instead of a probability distribution $\mathbf{x}$ where $\sum_v \mathbf{x}_v = 1$, we allow it to be any vector with nonnegative entries. A step of the "random walk" still means replacing $\mathbf{x}$ by $\tilde{\mathbf{W}}\mathbf{x}$, so after k steps, the vector becomes $\tilde{\mathbf{W}}^k \mathbf{x}$.

Let $s = \sum_v \mathbf{x}_v$. Note that the sum s is preserved after each step because $\tilde{\mathbf{W}}$ is a stochastic matrix. Therefore, $\mathbf{x}$ is just some probability distribution scaled by s . It also follows that this vector will converge to $s \cdot \pi$ where π is the stationary distribution, and it will take $T_{\text{mix}}(G, \varepsilon)$ time for the total variation distance to be $\leq s \cdot \varepsilon$.

Now, suppose M is a natural number such that, if we start from $\mathbf{e}$ concentrated on a single vertex and sum 1, then $d_{TV}(\tilde{\mathbf{W}}^M \mathbf{e}, \pi) \leq \varepsilon$. In other words, $T_{\text{mix}}(G, \varepsilon) \leq M$ if we restrict only to $\mathbf{x} = \text{some } \mathbf{e}$. We claim that $T_{\text{mix}}(G, \varepsilon) \leq M$ for the general $\mathbf{x}$ too.

Consider any starting distribution $\mathbf{x}$. Define vectors $\mathbf{e}^v$ for all vertices v , defined as follows:

$$\mathbf{e}_u^v = \begin{cases} \mathbf{x}_v & \text{if } v = u \\ 0 & \text{otherwise} \end{cases}$$

Intuitively, $\mathbf{e}^v$ is a vector that represents “the v component” of $\mathbf{x}$.

Note that $\mathbf{e}^v$ is a probability distribution scaled by $\mathbf{x}_v$. Thus, by definition of M and by the discussion at the start of the proof, we have

$$d_{TV}(\tilde{\mathbf{W}}^M \mathbf{e}^v, \mathbf{x}_v \pi) \leq \varepsilon \cdot \mathbf{x}_v$$

So after M steps, the total variation distance is $\leq \varepsilon \cdot \mathbf{x}_v$.

Now, each $\mathbf{e}^v$ can be treated independently; they don't interact with each other. They also run in parallel; a single step of the random walk advances all the $\mathbf{e}^v$ at once. Therefore, also after M steps, we have

$$d_{TV}(\tilde{\mathbf{W}}^M \mathbf{e}^v, \mathbf{x}_v \pi) \leq \varepsilon \cdot \mathbf{x}_v \quad \text{for all } v.$$

Now, the total variation distance in general is subadditive, i.e. it satisfies the following:

$$d_{TV}(\mathbf{x}_1 + \mathbf{x}_2, \mathbf{y}_1 + \mathbf{y}_2) \leq d_{TV}(\mathbf{x}_1, \mathbf{y}_1) + d_{TV}(\mathbf{x}_2, \mathbf{y}_2)$$

To see this, just apply the triangle inequality to each component of the vector.

Therefore, by adding over all the inequalities, we have

$$d_{TV}\left(\sum_v \tilde{\mathbf{W}}^M \mathbf{e}^v, \sum_v \mathbf{x}_v \pi\right) \leq \varepsilon \cdot \sum_v \mathbf{x}_v$$

But recall that $\mathbf{x}$ is a probability distribution, so $\sum_v \mathbf{x}_v = 1$. Moreover, by how $\mathbf{e}^v$ is defined, $\sum_v \mathbf{e}^v = \mathbf{x}$. Therefore

$$d_{TV}(\tilde{\mathbf{W}}^M \mathbf{x}, \pi) \leq \varepsilon$$

In other words, after M steps, the total variation distance to π is $\leq \varepsilon$. This is true for all $\mathbf{x}$, so it follows $T_{\text{mix}}(G, \varepsilon) \leq M$. This proves the claim.

3.2 Insight and discussion

The main insight is that the procedure of random walk can be applied to distributions that are not probability distributions too, and it basically just acts as a probability distribution with a constant scaling (that is preserved throughout the entire process). That lets you decompose the original distribution into each vertex. You know that within so many steps, the distribution that originated from a vertex will be distributed across the graph, so you can add up everything and make sure that your calculations are correct.

When I solved this problem, my intuition was using paint colors. Represent the probability distribution using paint from each vertex that adds up to 1 unit total. The paint from each vertex has a unique color and will not mix with any other color. Now, after long enough time, the paint from each color will be distributed across the graph, and you just take the sum of all paint of all colors. This gave me the idea to write down the solution as above.

4 Problem 4

Let $G = (V, E)$ be a graph with adjacency matrix $\mathbf{A}$. Prove that

$$\lambda_{\max}(\mathbf{A}) \geq 2 \cdot \max_H \frac{|E(H)|}{|V(H)|}$$

where H is taken over all non-null subgraphs (not necessarily induced) of G .

4.1 My solution

Use the equation for largest eigenvalue in lecture 4 slide 14. Let $R_{\mathbf{A}}$ be the Rayleigh quotient for the adjacency matrix $\mathbf{A}$:

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

We now investigate what this Rayleigh quotient means.

Let $\mathbf{x}$ be a 0-1 vector that is not all zero. This corresponds to a nonempty set $S \subseteq V$, specifically $S = \{v : \mathbf{x}_v = 1\}$.

The numerator $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ is the sum of entries in the form $\mathbf{x}_u \mathbf{A}_{uv} \mathbf{x}_v$ over all u, v . Since $\mathbf{A}$ is an adjacency matrix, it's 1 iff u, v are adjacent. Since $\mathbf{x}$ is a 0-1 vector, this term is 1 iff u, v are adjacent and u, v are both in S , and 0 otherwise. Therefore, the numerator is $2 \cdot |E(G[S])|$. (There's a factor of 2 since an edge uv is counted twice, as u, v and as v, u .)

The denominator $\mathbf{x}^\top \mathbf{x}$ is the sum of entries in the form $\mathbf{x}_v^2$. Since $\mathbf{x}$ is a 0-1 vector, this term is 1 iff $v \in S$, and 0 otherwise. Therefore, the denominator is $|V(G[S])|$.

In other words, for a 0-1 vector $\mathbf{x}$ represented by S ,

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{2 \cdot |E(G[S])|}{|V(G[S])|}$$

Now,

$$\lambda_{\max}(\mathbf{A}) = \max_{\mathbf{x}} R_{\mathbf{A}}(\mathbf{x}).$$

Therefore, in particular,

$$\lambda_{\max}(\mathbf{A}) \geq R_{\mathbf{A}}(\mathbf{x})$$

over all 0-1 vectors $\mathbf{x}$ that are not all zero. Substituting in the observation,

$$\lambda_{\max}(\mathbf{A}) \geq 2 \cdot \frac{|E(G[S])|}{|V(G[S])|}$$

for all nonempty set S .

It just remains to generalize this to the H in the problem, because the problem says H is not necessarily induced. But this is trivial. If H has vertex set S , then $|E(H)| \leq |E(G[S])|$, so H will not have more edges than $G[S]$. So H will not give a better bound than $G[S]$. This proves the claim.

4.2 Insight and discussion

Because we work with the adjacency matrix $\mathbf{A}$ instead of the normalized Laplacian, a whole ton of results from the lectures cannot be used. We don't have too many to work with; it's primarily the variational characterizations from the first half of lecture 4.

Considering this, we do know we have the bound for largest eigenvalue based on Rayleigh quotient:

$$\lambda_{\max}(\mathbf{A}) \geq R_{\mathbf{A}}(\mathbf{x})$$

for any $\mathbf{x}$. How can this relate to the problem?

Since the numerator $|E(H)|$ counts edges, and the adjacency matrix $\mathbf{A}$ involves edges, it's possible to add up entries of $\mathbf{A}$ to count up edges. This may also lead to the 0-1 vector observation: an entry $\mathbf{A}_{u,v}$ is counted iff u, v are present in the graph. To model whether a vertex is present or not, we may use a 0-1 vector for $\mathbf{x}$; then $\mathbf{A}_{u,v}$ is counted iff $\mathbf{x}_u = \mathbf{x}_v = 1$. And once you obtain this $\mathbf{x}$ idea, it's pretty easy to show it works.